

# DS701 Homework 4: Probability, MLE, and EM

Released Mon Sep 28 · due Mon Oct 5, 11:59 pm on Gradescope

## SOLUTIONS

### Problem 1: A screening test, three ways

*Practices: Bayes' rule with the base rate, and the odds / likelihood-ratio form.*

A screening test for a disease  $D$  has **sensitivity**  $P(+ | D) = 0.90$  (positive in 90% of people who have  $D$ ) and **specificity**  $P(- | \bar{D}) = 0.90$  (negative in 90% of people who do not). Bayes' rule for events, with the denominator expanded by the law of total probability (with  $C_k \in \{D, \bar{D}\}$ ):

$$P(D | +) = \frac{P(+ | D) P(D)}{P(+ | D) P(D) + P(+ | \bar{D}) P(\bar{D})}.$$

- (a) In the general population the **prevalence** (base rate) is  $P(D) = 0.01$ . A randomly chosen person tests positive. Compute  $P(D | +)$ , showing the two terms of the denominator separately. In one sentence, why is the answer so far below the 90% that “the test is 90% accurate” suggests?  $P(D | +) = \underline{\hspace{2cm}}$
- (b) The same test is used in a high-risk clinic where  $P(D) = 0.10$ . Recompute  $P(D | +)$ . Then state the practical lesson in one or two sentences: the test did not change — what did, and what does that say about using one test both for mass screening and in a specialist clinic?  $P(D | +) = \underline{\hspace{2cm}}$
- (c) **Odds form.** Define the odds of an event as  $O(D) = P(D)/P(\bar{D})$  and the positive likelihood ratio as  $LR_+ = P(+ | D)/P(+ | \bar{D})$ . Show, starting from Bayes' rule written once for  $D$  and once for  $\bar{D}$ , that

$$O(D | +) = LR_+ \times O(D)$$

(posterior odds = likelihood ratio  $\times$  prior odds — no denominator needed). Optionally, then compute  $LR_+$  for this test, redo (a) and (b) in odds and convert back to probabilities to confirm your answers, and finally: a person from the general population tests positive on **two independent** administrations of the test — what is  $P(D | ++)$ ? (Independence here means the second test multiplies the odds by  $LR_+$  again.)  $LR_+ = \underline{\hspace{2cm}}$   $P(D | ++)$  =  $\underline{\hspace{2cm}}$

### Solution

- (a)  $P(+ | D)P(D) = 0.9 \times 0.01 = 0.009$ ;  $P(+ | \bar{D})P(\bar{D}) = 0.1 \times 0.99 = 0.099$ ;  $P(+)$  = 0.108. So  $P(D | +) = 0.009/0.108 = 1/12 \approx 0.083$ . The false positives (0.099) outnumber the true positives (0.009) eleven to one because the disease is rare: 10% of a large healthy

population is far more people than 90% of a tiny sick one. “90% accurate” describes  $P(+ | D)$ , not  $P(D | +)$  — the base rate is what separates them.

- (b)  $0.9 \times 0.1 = 0.09$ ;  $0.1 \times 0.9 = 0.09$ ;  $P(D | +) = 0.09/0.18 = 0.5$ . Only the prior changed, from 0.01 to 0.10, and the posterior went from 0.083 to 0.5. A positive result means very different things in the two settings: in mass screening it is mostly a false alarm and should trigger a confirmatory test, not a diagnosis; in the clinic, where the prior is high, the same result is a coin flip and worth acting on. (Common mistake: reporting 0.9 in both cases — that ignores the base rate entirely.)
- (c)  $P(D | +) = P(+ | D)P(D)/P(+)$  and  $P(\bar{D} | +) = P(+ | \bar{D})P(\bar{D})/P(+)$ . Divide the first by the second;  $P(+)$  cancels:  $O(D | +) = \frac{P(+ | D)}{P(+ | \bar{D})} \cdot \frac{P(D)}{P(\bar{D})} = LR_+ \times O(D)$ . Here  $LR_+ = 0.9/0.1 = 9$ . (a): prior odds  $0.01/0.99 = 1:99$ ; posterior odds  $9:99 = 1:11$ ;  $P = 1/(1 + 11) = 1/12 \approx 0.083$ . (b): prior odds  $1:9$ ; posterior odds  $9:9 = 1:1$ ;  $P = 0.5$ . Two independent positives: odds  $1:99 \times 9 \times 9 = 81:99$ ;  $P(D | ++)= 81/180 = 0.45$ . Each positive multiplies the odds by 9, but a 1:99 start needs two of them just to reach even odds — the same base-rate effect, now in a form where repeated evidence is a simple product.

## Problem 2: Maximum likelihood, from scratch

*Practices: the MLE recipe — likelihood, log, derivative, solve — on two models, and knowing why the log is there.*

- (a) **Bernoulli.** Each observation is  $x_n \in \{0, 1\}$  with  $P(x_n = 1) = p$ , so  $P(x_n | p) = p^{x_n}(1 - p)^{1-x_n}$ . For  $N$  i.i.d. observations with  $k = \sum_n x_n$  ones: write the likelihood  $L(p)$ , the log-likelihood  $\ell(p)$ , set  $d\ell/dp = 0$ , and solve for  $\hat{p}$ . Then evaluate  $\hat{p}$  for the data 1, 0, 1, 1, 0, 1, 1, 1, 0, 1. (Say in one line why the critical point is a maximum and not a minimum.) For the data,  $\hat{p} = \underline{\hspace{2cm}}$
- (b) **Gaussian mean and variance.** Start from the Gaussian log-likelihood

$$\ell(\mu, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2.$$

Derive  $\hat{\mu}$  from  $\partial\ell/\partial\mu = 0$ . Then treat  $v = \sigma^2$  as the variable, derive  $\hat{\sigma}^2$  from  $\partial\ell/\partial v = 0$  (substituting  $\hat{\mu}$ ), and evaluate both on the data 2, 3, 5, 6, 9. Compare  $\hat{\sigma}^2$  with the sample variance  $s^2$  that divides by  $N - 1$  — which is larger, and by what factor here?  $\hat{\mu} = \underline{\hspace{2cm}}$   
 $\hat{\sigma}^2 = \underline{\hspace{2cm}}$   $s^2 = \underline{\hspace{2cm}}$  factor:  $\underline{\hspace{2cm}}$

- (c) **Why the log?** Give the two reasons the lecture gives for maximizing  $\ell$  rather than  $L$ : one about *where* the maximum is, one about the *form* of the function. Then add the practical one: for  $N = 2000$  observations from (a), every factor of  $L$  is at most 1 — say roughly how small  $L$  is at best (e.g. compare  $0.5^{2000}$  with the smallest positive double, about  $10^{-308}$ ) and why  $\ell$  does not have this problem.

**Solution**

- (a)  $L(p) = \prod_n p^{x_n} (1-p)^{1-x_n} = p^k (1-p)^{N-k}$ ;  $\ell(p) = k \log p + (N-k) \log(1-p)$ ;  $\frac{d\ell}{dp} = \frac{k}{p} - \frac{N-k}{1-p} = 0 \Rightarrow k(1-p) = (N-k)p \Rightarrow \hat{p} = k/N$ . It is a maximum because  $\ell'' = -k/p^2 - (N-k)/(1-p)^2 < 0$  (or:  $\ell \rightarrow -\infty$  at both ends  $p \rightarrow 0, 1$ , so the single critical point is the peak). Data:  $k = 7$ ,  $N = 10$ ,  $\hat{p} = 0.7$ .
- (b)  $\partial \ell / \partial \mu = \frac{1}{\sigma^2} \sum_n (x_n - \mu) = 0 \Rightarrow \hat{\mu} = \frac{1}{N} \sum_n x_n$  (matches the lecture's one-line derivation). With  $v = \sigma^2$ :  $\ell = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log v - \frac{1}{2v} S$ , where  $S = \sum_n (x_n - \hat{\mu})^2$ ;  $\partial \ell / \partial v = -\frac{N}{2v} + \frac{S}{2v^2} = 0 \Rightarrow \hat{v} = S/N$ , i.e.  $\hat{\sigma}^2 = \frac{1}{N} \sum_n (x_n - \hat{\mu})^2$ . Data:  $\hat{\mu} = 25/5 = 5$ ; deviations  $-3, -2, 0, 1, 4$ , squares  $9, 4, 0, 1, 16$ ,  $S = 30$ ;  $\hat{\sigma}^2 = 30/5 = 6$ ;  $s^2 = 30/4 = 7.5$ . The  $N-1$  version is larger by the factor  $N/(N-1) = 5/4$ .
- (c) (i) log is monotonically increasing, so  $\ell$  and  $L$  have their maximum at the *same*  $\theta$  — nothing is lost. (ii) The log turns the product over observations into a sum, so the derivative is a sum of simple terms and the algebra (as in (a) and (b)) is short. (iii) Numerically:  $L$  is a product of thousands of numbers  $\leq 1$ ; even in the best case  $0.5^{2000} = 10^{-602}$ , far below  $10^{-308}$ , so in floating point  $L$  underflows to exactly 0 for every  $p$  and cannot be maximized.  $\ell = 2000 \log 0.5 \approx -1386$  is an ordinary number; sums of logs stay in range.

**Problem 3: One EM iteration by hand**

*Practices: the E-step and M-step formulas on numbers small enough to see what they do, and the argument for why EM behaves.*

A one-dimensional mixture of  $K = 2$  Gaussians, both with variance  $\sigma^2 = 1$  (**keep the variances fixed at 1 throughout** — update only means and weights). Current parameters:  $\mu_1 = 0$ ,  $\mu_2 = 2$ ,  $\pi_1 = \pi_2 = 0.5$ . Data:  $x_1 = 0$ ,  $x_2 = 1$ ,  $x_3 = 2$ ,  $x_4 = 4$ . Useful values:  $e^{-0.5} \approx 0.607$ ,  $e^{-2} \approx 0.135$ ,  $e^{-8} \approx 0.00034$ ; note that with equal variances the factor  $1/\sqrt{2\pi}$  is the same in every term and cancels in the responsibilities.

- (a) **E-step.** For each data point compute the responsibilities  $\gamma_{i1}, \gamma_{i2}$  (3 decimals). Before computing, say for which point the answer must be exactly 0.5 and why. ( $\gamma_{i1}, \gamma_{i2}$ ) for  $x_1, \dots, x_4$ :
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- (b) **M-step.** Compute  $N_1, N_2$ , the new means  $\mu'_1, \mu'_2$  and the new weights  $\pi'_1, \pi'_2$ . In one sentence each: why did  $\mu_2$  move more than  $\mu_1$ , and why did the weights become unequal?  $N_1 =$  \_\_\_\_\_  $N_2 =$  \_\_\_\_\_  $\mu'_1 =$  \_\_\_\_\_  $\mu'_2 =$  \_\_\_\_\_  $\pi'_1 =$  \_\_\_\_\_  $\pi'_2 =$  \_\_\_\_\_
- (c) The lecture states that the log-likelihood  $\ell(\theta)$  never decreases from one EM iteration to the next. Explain the mechanism in words: what the E-step computes, what the M-step maximizes and why that has a closed form, and why alternating the two cannot lower  $\ell$ . Then say what this guarantees about the sequence  $\ell^{(t)}$ , what it does **not** guarantee, and what is done about that in practice (relate to Homework 3, Problem 1).
- (d) **Hard vs. soft.** Without recomputing, describe what the four responsibilities in (a) would look like if both variances were  $\sigma^2 = 0.01$  instead of 1, and what the M-step for the means would then reduce to. Which algorithm is that? State the one practical difference and the

one conceptual difference between GMM/EM and  $k$ -means that the lecture draws.

### Solution

- (a) With equal  $\pi_k$  and equal variances,  $\gamma_{i1} = \frac{e^{-(x_i-0)^2/2}}{e^{-(x_i-0)^2/2} + e^{-(x_i-2)^2/2}}$ . The point  $x_2 = 1$  is equidistant from the two means, so both numerators are  $e^{-0.5}$  and  $\gamma_{21} = \gamma_{22} = 0.5$  exactly.  $x_1 = 0$ :  $e^0 = 1$  vs.  $e^{-2} = 0.135$ :  $\gamma_{11} = 1/1.135 = 0.881$ ,  $\gamma_{12} = 0.119$ .  $x_2 = 1$ : 0.5, 0.5.  $x_3 = 2$ :  $e^{-2}$  vs.  $e^0$ :  $\gamma_{31} = 0.119$ ,  $\gamma_{32} = 0.881$ .  $x_4 = 4$ :  $e^{-8} = 0.00034$  vs.  $e^{-2} = 0.135$ :  $\gamma_{41} = 0.00034/0.13534 = 0.0025$  (call it 0.002–0.003),  $\gamma_{42} = 0.9975$ . Each row sums to 1. (Common mistake: forgetting the square in the exponent, or using  $e^{-(x-\mu)^2}$  instead of  $e^{-(x-\mu)^2/2}$ .)
- (b)  $N_1 = 0.881 + 0.5 + 0.119 + 0.0025 = 1.5025 \approx 1.503$ ;  $N_2 = 0.119 + 0.5 + 0.881 + 0.9975 = 2.4975 \approx 2.498$  (they sum to  $n = 4$ ).  $\mu'_1 = (0.881 \cdot 0 + 0.5 \cdot 1 + 0.119 \cdot 2 + 0.0025 \cdot 4)/1.5025 = 0.748/1.5025 \approx 0.498$ .  $\mu'_2 = (0.119 \cdot 0 + 0.5 \cdot 1 + 0.881 \cdot 2 + 0.9975 \cdot 4)/2.4975 = 6.252/2.4975 \approx 2.503$ .  $\pi'_1 = 1.5025/4 \approx 0.376$ ,  $\pi'_2 = 2.4975/4 \approx 0.624$ .  $\mu_2$  moved more because the far point  $x_4 = 4$  is almost entirely component 2's responsibility and drags its weighted mean right, while component 1 only picks up the half-share of  $x_2 = 1$ . The weights became unequal because component 2 now “owns” more total responsibility ( $N_2 > N_1$ ) — three of the four points lie at or to the right of the midpoint 1. (Check: the log-likelihood rises from  $-7.999$  to  $-7.315$ .)
- (c) The E-step computes, for the current  $\theta$ , the posterior probability  $\gamma_{ik} = P(z_i = k \mid x_i, \theta)$  that each point came from each component — Bayes' rule with  $\pi_k$  as prior and  $\mathcal{N}(x_i \mid \mu_k, \Sigma_k)$  as likelihood. Given those soft assignments, the M-step maximizes the *expected complete-data* log-likelihood  $\sum_i \sum_k \gamma_{ik} [\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)]$ ; because the log now sits on individual Gaussians (not on a sum of them), the derivatives separate and the maximizers are the weighted MLE formulas — the closed form. Why  $\ell$  cannot go down: with the responsibilities fixed at their E-step values, that expected complete-data objective (plus a term that does not depend on  $\theta$ ) is a lower bound on  $\ell(\theta)$  that *touches*  $\ell$  at the current  $\theta$ ; the M-step raises the bound, and  $\ell$  sits above the raised bound, so  $\ell(\theta^{\text{new}}) \geq \ell(\theta^{\text{old}})$  (the lecture asserts the monotone increase; the “touching lower bound” sentence is the reason). Guarantee:  $\ell^{(t)}$  is non-decreasing and bounded above, so it converges — which is what the log-likelihood-change stopping rule relies on. Not guaranteed: that the limit is the global maximum; EM finds a *local* maximum and depends on the initialization — exactly like the  $k$ -means run in HW3 Problem 1(c) that stopped at WCSS 74 instead of 33.5. In practice: initialize from  $k$ -means and/or run several restarts, keep the highest  $\ell$ .
- (d) With  $\sigma^2 = 0.01$  the exponents are 100 times larger, so every responsibility saturates:  $x_1$  is  $\approx (1, 0)$ ,  $x_3$  and  $x_4$  are  $\approx (0, 1)$ , and only the exactly-equidistant  $x_2 = 1$  stays at  $(0.5, 0.5)$  (a knife-edge tie). With 0/1 responsibilities,  $\mu'_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i$  is just the plain mean of the points assigned to component  $k$  — the  $k$ -means update; the E-step is the nearest-center assignment. So EM with tiny, equal, fixed variances *is* Lloyd's algorithm. Practical difference (lecture): GMM makes soft assignments (“soft  $k$ -means”) where  $k$ -means makes hard ones. Conceptual difference: the GMM states an explicit generative assumption — data drawn from a mixture of Gaussians with their own covariances — which can be examined and replaced;  $k$ -means only declared an objective, and its implicit spherical/equal-variance assumption was never stated.